
CS204: Algorithms & Data Structures Lab

Week # 07 (1 Questions, 100 Marks)

Lab session: AL1

Held on: 09-Sep-2024 (Mon)

Lab Timings: 14:00 to 17:00 Hours Pages: 3

Submission time: 16:45 Hrs, 09-Sep-2024

Instructors Dr. V. Vijaya Saradhi

Head TAs Adithya K Moorthy & Laxita Agrawal

Department of CSE, IIT Guwahati

Question 1: (100 points)

Templates and sorting.

Task 01 Create a class named `Ratio` with the following specifications

1. (2 marks) Two integers named `numerator`, `denominator` as private members
2. public member functions
 - (a) (2 marks) A constructor with the following specification
 - **Input argument 1** An int numerator with default value 0
 - **Input argument 2** An int denominator with default value 1
 - **Functionality** Initialize the numerator and denominator.
 - (b) (2 marks) A copy constructor with the following specification
 - **Input argument 1** A const `Ratio` object.
 - **Functionality** This should copy the contents of input argument 1 in this constructor.
 - (c) (2 marks) An overloaded `<` operator with the following specification
 - **Input argument 1:** const reference to `Ratio` object
 - **Returns:** bool
 - **Functionality:** compare the two `Ratio` objects and return true if LHS ratio is less than RHS ratio. False otherwise.
 - (d) (2 marks) An overloaded `>>` operator with the following specification
 - **Input argument 1:** `istream` reference
 - **Input argument 2:** const `Ratio` object
 - **Returns:** `istream` reference
 - **Functionality:** Read the numerator and denominator values from the input file (no need to use file I/O) into the given object
 - (e) (2 marks) An overloaded `<<` operator with the following specification
 - i. **Input argument 1:** `ostream` reference
 - ii. **Input argument 2:** const `Ratio` object
 - iii. **Returns:** `ostream` reference
 - iv. **Functionality:** Print to standard terminal the `Ratio` as “1/3”, “2/7” etc.

- (f) (2 marks) An overloaded `[]` operator with the following specification
 - i. **Input argument 1:** `int` data type
 - ii. **Returns:** pointer to `Ratio`
 - iii. **Functionality:** Performs array element assignment with `Ratio` object

Task 02 Create a class template named `sort` with the following specifications

1. (3 marks) Private members
 - (a) An `int` representing number of elements
 - (b) A variable named `data` of type a pointer to a template typename `T` to store data.
2. Public members
 - (a) (2 marks) A constructor with the following specification
 - **Input argument 1** An `int` with default value of 10 representing number of elements.
 - **Functionality** Initialize the memory for the `data`.
 - (b) (2 marks) A constructor with the following specification
 - **Input argument 1** A `const` pointer to pointer to template typename `T`
 - **Input argument 2** A `int` representing number of elements
 - **Functionality** Initialize the data members `number of elements` and `data`. Copy the contents of the input arguments to data members.
 - (c) (2 marks) A copy constructor with the following specification
 - **Input argument 1** A `const` pointer to template typename `T`.
 - **Functionality** This should initialize the memory for the `data`. Copy the contents of input argument 1 in this constructor.
 - (d) (49 marks) Declare a `const` function named `insertion_sort` with the
 - **Input:** default input `bool` argument `TRUE` representing type of sorting. `TRUE`: represent ascending order. `FALSE`: represent descending order.
 - **Returns:** `void`
 - **Functionality:** Sort the data using insertion sort algorithm.
 - (e) (2 marks) Declare a function named `t_min` with the following specifications.
 - **Input argument 1:** `const` template typename `T` `a`
 - **Input argument 2:** `const` template typename `T` `b`
 - **Output:** `bool`
 - **Functionality:** Return the minimum of (`a`, `b`)
 - (f) (2 marks) An overloaded `>>` operator with the following specification
 - **Input argument 1:** `istream` reference
 - **Input argument 2:** a pointer to template typename `T`
 - **Returns:** `istream` reference
 - **Functionality:** Read the values from the input file (no need to use file I/O) into `data`

- (g) (2 marks) An overloaded `<<` operator with the following specification
- **Input argument 1:** ostream reference
 - **Input argument 2:** const pointer to template typename T object
 - **Returns:** ostream reference
 - **Functionality:** Print to standard terminal the values in the data.
- (h) (2 marks) A destructor with the following specifications.
- **Input arguments:** None
 - **Returns:** None
 - **Functionality:** Free the dynamically allocated memory of the data member.

Task 04 (20 marks) In the main function perform the following

1. Input file format: First line contains number of elements to sort
2. Data follows the first line.
3. Read integer data from input file `int-sort.txt`
4. (2 marks) Sort data in ascending order
5. (2 marks) Sort data in descending order

6. Read character data from input file `char-sort.txt`
7. (2 marks) Sort data in ascending order
8. (2 marks) Sort data in descending order

9. Read float data from input file `float-sort.txt`
10. (2 marks) Sort data in ascending order
11. (2 marks) Sort data in descending order

12. Read string data from input file `string-sort.txt`
13. (2 marks) Sort data in ascending order
14. (2 marks) Sort data in descending order

15. Read Ratio data from input file `ratio-sort.txt`
16. (2 marks) Sort data in ascending order
17. (2 marks) Sort data in descending order